Christian Tabbah

christiantabbah@mail.utoronto.ca | https://github.com/ChrisJTab

Education

University of Toronto **September 2020 - April 2024**

HBSc in Computer Science, Information Security Specialist and Mathematics Minor - CGPA: 3.74/4.0

University of Toronto **September 2024 - Ongoing**

MSc in Computer Science, expected completion in October 2026

Experience

Full Stack Software Developer - VeritableSoft **January 2021 - August 2023**

- Used the .NET framework, C# and SQL to create and improve pages for the company's website, including the backend and database implementations
- Worked on migrating the application from Bootstrap 2 to Bootstrap 4

Teaching Assistant - University of Toronto **January 2023 - Ongoing**

- **Courses taught:**
  - CSC369H5 (Operating Systems) - Fall 2023, 2024, 2025 / Winter 2026
  - CSC347H5 (Introduction to Information Security) - Fall 2023 / 2024
  - CSC148H5 (Introduction to Computer Science) - Winter 2023 / Summer 2025
  - CSC427H5 (Computer Security) - Winter 2024
  - CSC363H5 (Computational Complexity and Computability) - Winter 2024
  - CSC343H1 (Introduction to Databases) - Fall 2025
- Plan and implement lesson plans for tutorials or practicals.
- Provide and create resources for students, and answer student questions on discussion boards.
- Hold office hours and mark students' tests, quizzes, assignments and exams.
- Lecture while the professor is at a conference (CSC347H5).

Summer Camp Counselor - Kids4Kids **June 2018 - August 2022**

- Supervised/guided groups of children, ensuring their safety and engagement in daily activities, fostering teamwork among campers. Successfully created and organized activities for over 250 campers at a time.
- Worked 1-1 with special needs campers to create a safe environment for them within the larger camp.

Computer Forensics Research Project - University of Toronto **April 2023 - August 2023**

- Supervised by Dr. Andi Bergen.
- Researched and wrote a paper on methods of data loss and recovery at the lowest levels of software and hardware. Tested methods of data recovery on different operating systems. Also researched different operating systems and their respective file system architectures.

Master's Research Project - University of Toronto **September 2024 - Ongoing**

- Supervised by Dr. Angela Demke Brown & Dr. Ashvin Goel.
- Build a GPU-accelerated transactional database that operates beyond VRAM limits by minimizing data movement and dynamically orchestrating CPU-GPU work while preserving determinism and OLTP correctness.
- Engineer an end-to-end GPU staging & eviction layer consisting of device free-list ring, epoch-aware hot-cache, selective dirty write-back, and scatter–gather DMA. The eviction layer turns multi-million-key OLTP batches into a predictable, low-latency pipeline.
- Add profiling and reproducible tooling (logs→CSV→TikZ) to guide optimization.

Projects

File System and MySH | C

- Created an ext2 file system using C and implemented "cd", "mkdir", "cp", "rm", hard links and soft links.
- Created a shell using C and implemented the "echo", "wc", piping("|"), "grep", "cd", forking processes, backgrounding processes and networking support.

Parcube | React Native, TypeScript, HTML, Firestore, Google API, GitHub, Agile

- Using Agile methodology to work as a team in multiple sprints, we created a fully functional app that allows users to rent out their parking spots to others, as well as rent other people's parking spots.
- The application also parses through already existing parking spots and displays them on a map using Google API.
- Implemented reward systems to reinforce user commitment to the application.

ML Challenge | Python, PyTorch, Pandas, NumPy

- Built a machine learning model to classify survey answers using a mix of Naive Bayes and Decision Trees.
- This model produced the best results compared to the other groups participating, with a test accuracy of 85-90%.

UDP Data Transfer Implementation | Python

- Implemented a high-performance UDP-based data transfer system resulting in reliable data transfer.

Distributed System URLShortener | Java, SQL

- Used Java to implement a scalable, fault-tolerant URL shortening server with high availability and consistency.
- Created a reverse proxy that utilized consistent hashing, a distributed database (sharding) and had full monitoring capabilities.

Distributed System URLShortener | Docker Swarm, Cassandra, Redis, Python

- Used Docker Swarm, Cassandra, Redis and Python to implement a scalable, fault-tolerant URL shortening server with high availability and consistency.

RocksDB-Hailstorm Disaggregation Prototype | Scala, C++

- Prototyped RocksDB disaggregated storage using the Hailstorm architecture; modified RocksDB to support a remote-compaction/offload workflow and documented system design + integration approach.
- Built an end-to-end experimental stack (Hailstorm + RocksDB + YCSB) to evaluate LSM-tree workloads (e.g., uniform/Zipfian), and produced a project status report summarizing methodology and progress.
- Identified and documented a critical stability blocker ("bad magic table number") causing crashes when running RocksDB on the Hailstorm mount path; compiled debugging notes and proposed next-step investigation to unblock benchmarking.

Exploring On-Path Processing for Efficient Data Intensive Workloads on GPU clusters | C++, CUDA

- Built a distributed on-path processing prototype for GPU-accelerated OLAP DISTINCT, where two multi-GPU worker hosts perform local + CPU-side deduplication before RDMA-sending reduced results to a final aggregation host, which completes global DISTINCT on the destination GPU via CUB radix sort + unique.
- Implemented a UCX-based RDMA client/server data path with zero-copy writes into pre-registered server buffers, plus threshold-triggered pipelining and queue-based asynchronous RDMA writes to overlap CPU processing with network I/O.
- Engineered the CPU dedup stage using an adapted concurrent hashmap optimized for lookup-heavy workloads (per-bucket std::map), and coordinated receiver-side progress using a lightweight counter-based synchronization protocol.

Outreach

President of the UofT Mississauga Poker Club — **August 2023 - September 2024**

- Founded and scaled the club to 200+ active members through recurring events and community-building initiatives.
- Organized biweekly tournaments and led beginner-to-intermediate workshops focused on probability, expected value (EV), combinatorics, risk management, and decision-making under uncertainty.

Vice President of Social Coordination of the UofT Computer Science Graduate Student Union

September 2025 - Ongoing

- Organize large-scale events for graduate students and coordinate logistics across executive team members.
- Lead event planning and communications to maintain consistent student engagement across the academic year.

Skills

Languages: C, C++, CUDA, Python, Bash, SQL, Java, JavaScript, TypeScript, CSS, HTML, Assembly, C#
Frameworks: React, Bootstrap, Django, .NET Framework, Node.js, ASP.NET, jQuery, Tailwind
Tools: Git, Docker, AWS, Azure, Firestore, Postgres, PHP, Neo4j, MongoDB, Cassandra, Redis, Nginx
Bilingual: French (Fluent), English (Fluent), Armenian (Rusty)

Personal Achievements

University of Toronto Dean's List Scholar

- Awarded to students who score at least a CGPA of 3.5 or higher after each quarter of their degree
- Achieved four times during undergraduate degree

Black Belt in Taekwondo

- Worked hard for 7 years to achieve a black belt in Taekwondo
- Soon to earn my Second Degree black belt.